Lesson notes · for you

How a classifier decides

AI · Level 3 · Lesson 2

Time	Materials
25 min	Squared paper (or a grid you draw) and a pencil; Ten fruits of one kind, at home or at a market stall (ask first); A ruler

Learning goals

- Describe each example by features: numbers you can measure.
- Classify a new example by its nearest neighbours, and see how the number of neighbours changes the answer.
- Explain why the choice of features matters.

Activity: Neighbours on paper

1. Measure (5 min): Collect eight leaves from two kinds of tree (or draw eight shapes in two groups). Measure each leaf's length and width in centimetres and write its label: tree A or tree B.
2. Chart (5 min): On squared paper, draw a chart with length along the bottom and width up the side. Mark tree A leaves as filled dots and tree B leaves as open dots.
3. Mystery (5 min): A partner secretly picks a new leaf, measures it, and tells you only its length and width. Mark it on the chart as a triangle.
4. Vote (5 min): Draw lines to its 1, 3 and 5 nearest dots, and write what each k says. Then your partner reveals the real tree. Which k was right?
5. Swap features (5 min): Replace width with a poor feature, such as how dusty the leaf is. Do the two kinds still separate? Write one sentence on why features matter.

Works in pairs or at home. Any two kinds of objects will do: leaves from two trees, two kinds of beans, or two shapes you draw. Use a ruler for every measurement.

Discussion

- Why might k = 1 give a strange answer?
- Which two features would you use to sort bananas into ripe and not ripe?
- Can a classifier be better than the features it is given? Why?

Check yourself

- One feature I would measure for a classifier of my own, and why.

Try it this week

Score ten fruits of one kind from 0 to 10 for size and for softness, mark each on a chart on squared paper, and label it ripe or not. Then score a new fruit and let its three nearest neighbours vote. Taste it or cut it open: were they right?

Worksheet

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Name _____ Class _____ Date _____

1 **Your model must tell handwritten sevens from ones. Which feature helps more?**

◯ The writer's name ◯ How wide the digit is ◯ The colour of the ink

2 **Put the steps of classifying a new mango by its nearest neighbours in order.**

☐ Place it on the chart with the checked mangoes

☐ Find the k closest dots

☐ Measure its features: softness and colour

☐ Give it the label most of them have

☐ Count the labels of those neighbours

3 **With k = 1, a single mislabelled example can change the answer for new mangoes near it.**

◯ True ◯ False

4 **Match each classifier to a useful feature.**

Ripe or unripe mango	• • • •	How much of the leaf is spotted
Handwritten 7 or 1	• • • •	How high or low the voice is
Healthy or sick leaf	• • • •	Softness, from 0 to 10
Child's or adult's voice	• • • •	How wide the digit is

✎ **Think of something you could sort with a classifier. Which two features would you measure, and why those?**

Check your answers

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1 **How wide the digit is**

A seven is wider than a one, whoever writes it and in any ink.

2 **Top to bottom: 2, 3, 1, 5, 4**

You need its features before it has a place on the chart.

3 **True**

With k = 1 the closest example decides alone, so one wrong label misleads every new mango that lands next to it.

4 **Ripe or unripe mango — Softness, from 0 to 10**
Handwritten 7 or 1 — How wide the digit is
Healthy or sick leaf — How much of the leaf is spotted
Child's or adult's voice — How high or low the voice is

A useful feature changes between the classes and can be measured the same way every time.